

15. Today is Aune's fifteenth birthday. Her parents measure her height ("for the last time!") to be 62.5 inches. On her fourteenth birthday her height was 61.0 inches. They noted her height in inches has been increasing geometrically for the past five years. Determine Aune's height, in inches, on her tenth birthday. Report your answer as a decimal rounded to the nearest quarter of an inch.
16. Let $f(x) = 3 + 4\cos(x + \pi)$ with x measured in radians. Let k represent the amplitude of the graph of $f(x)$ and w represent the y-intercept of the graph of $f(x)$. Determine the sum $(k + w)$.
17. In $\triangle ABC$, D lies on $\overline{AC}$ such that $\overline{BD}$ bisects $\angle ABC$. $AB = 18$, $BC = 12$, and $DC = 7$. Determine the degree measure, in degrees, for the smallest angle of $\triangle ABC$. Express your answer as a decimal rounded to the nearest thousandth.
18. Let $\sin \theta = -\frac{1}{2}$ where $0 \leq \theta < 2\pi$ is measured in radians. The sum of all possible value(s) for θ is $k\pi$. Determine the value of k . Express your answer as an integer or as a common or improper fraction reduced to lowest terms.
19. When $2\sin(2x)\cos x - 4\cos^2 x = \sin x - 1$ is solved over $0 \leq x \leq 2\pi$, the sum of the solutions is $k\pi$. Determine the exact value of k . Express your answer as an integer or as a common or improper fraction reduced to lowest terms.
20. The equation of the directrix of a parabola with vertex $(-3, 2)$ that passes through $(9, -16)$ and opens down is either $x = k$ or $y = w$. Determine the equation of this directrix. Express your answer as the equation $x = k$ or $y = w$.